

Replication of Gourio & Miao (2010): “Firm Heterogeneity and the Long-Run Effects of Dividend Tax Reform”

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1 Introduction

This document replicates the main quantitative results of Gourio and Miao (2010, *AEJ: Macro*). The model is a Bewley-style stationary equilibrium of heterogeneous firms with idiosyncratic productivity shocks, capital adjustment costs, corporate and personal taxation, and a constraint that firms cannot repurchase shares. The exercise is to compute the long-run effects of cutting the dividend tax (and the joint cut in dividend and capital-gains taxes) on aggregate capital, output, consumption, and welfare.

All tables are generated by `GourioMiao2010.m` and live under `./SavedOutput/LatexInputs/`; figures reproduced from the original article are placed under `./SavedOutput/Graphs/` as image files.

2 Model

A representative household owns equity in a continuum of firms indexed by idiosyncratic productivity z . Each firm operates a Cobb–Douglas technology $y = zk^{\alpha_k} \ell^{\alpha_\ell}$, hires labor in a spot market at wage w , and chooses next-period capital k' . Capital adjustment costs are quadratic in net investment:

$$\Phi(k', k) = \frac{\phi^c}{2} \frac{(k' - (1 - \delta)k)^2}{k}.$$

Corporate income tax τ_{corp} is levied on profits net of depreciation expensing and a fraction ϕ of adjustment costs:

$$T = \pi - \delta k - \phi \Phi(k', k), \quad \pi = y - w\ell.$$

After-tax cash flow is paid out as dividends $d \geq 0$ or by issuing new equity $s \geq 0$; share repurchases are not allowed. Personal income, dividend, and capital-gains taxes are τ_i , τ_d , and τ_{cg} respectively. The household has preferences $U(C, L) = \log C - \frac{h}{2} L^2$, which delivers the labor-market clearing condition $hCL = (1 - \tau_i)w$. A no-arbitrage condition on equity returns pins the firm discount factor:

$$\hat{\beta} = \frac{1}{1 + r(1 - \tau_i)/(1 - \tau_{cg})}, \quad \beta(r(1 - \tau_i) + 1) = 1.$$

Firm problem (as written in GM2010). With d and k' both as decisions, the firm chooses (d, k') to maximize

$$F = \frac{1 - \tau_d}{1 - \tau_{cg}} d - s, \quad s = d + I + \Phi(k', k) - (\pi - \tau_{corp} T), \quad I \equiv k' - (1 - \delta)k,$$

subject to $d \geq 0$ and $s \geq 0$, and the Bellman equation $V(k, z) = \max_{d, k'} \{F + \hat{\beta} \mathbb{E}[V(k', z') \mid z]\}$.

3 Algebraic simplifications used in the code

Two static substitutions reduce the firm problem to a one-control Bellman equation in k . Both substitutions are exact and hold for any wage w (in or out of general equilibrium); they merely use the firm's interim FOCs analytically rather than numerically.

3.1 Substitute out labor: $\pi = (1 - \alpha_l) y$

The static labor FOC gives $w = \alpha_l z k^{\alpha_k} \ell^{\alpha_l - 1}$ and therefore $w \ell = \alpha_l y$. Profits are then

$$\pi = y - w \ell = (1 - \alpha_l) y.$$

Solving for the optimal ℓ and substituting back yields closed forms for output, labor demand, and profit in $(k, z; w)$:

$$\ell^*(k, z; w) = \left(\frac{\alpha_l z k^{\alpha_k}}{w} \right)^{1/(1 - \alpha_l)}, \quad y(k, z; w) = (z k^{\alpha_k})^{1/(1 - \alpha_l)} (\alpha_l / w)^{\alpha_l / (1 - \alpha_l)}, \quad \pi(k, z; w) = (1 - \alpha_l) y(k, z; w).$$

This identity follows from *firm* optimization at the labor stage; it does not require labor-market clearing, so it is valid both in and out of GE.

3.2 Eliminate the dividend as a decision

Use the budget constraint $s = d - A$ with

$$A(k', k, z; w, \theta) \equiv \pi - \tau_{corp} T - I - \Phi(k', k)$$

(the dividend the firm would pay if it issued no equity). Substituting into F :

$$F = s \left[\frac{1 - \tau_d}{1 - \tau_{cg}} - 1 \right] + \frac{1 - \tau_d}{1 - \tau_{cg}} A.$$

With $\tau_d \geq \tau_{cg}$ (baseline: $\tau_d = 0.25$, $\tau_{cg} = 0.20$), the coefficient on s is non-positive, so the firm sets s as small as feasible. Combined with $d \geq 0$:

- If $A \geq 0$: optimum is $d^* = A$, $s^* = 0$ and $F = \frac{1 - \tau_d}{1 - \tau_{cg}} A$.
- If $A < 0$: optimum is $d^* = 0$, $s^* = -A > 0$ and $F = A$.

In one line,

$$F(k', k, z; w, \theta) = \frac{1 - \tau_d}{1 - \tau_{cg}} \max\{A, 0\} + \min\{A, 0\},$$

with $d^*(k', k, z) = \max\{A, 0\}$ and $s^*(k', k, z) = \max\{-A, 0\}$. The dividend is therefore not an independent control: given any policy for k' , the dividend and equity-issuance policies are pinned down algebraically.

3.3 The simplified Bellman equation

After both substitutions, the firm's recursive problem is a single Bellman equation in k with the single control k' :

$$V(k, z) = \max_{k' \in \mathcal{K}} \left\{ \frac{1-\tau_d}{1-\tau_{cg}} \max\{A, 0\} + \min\{A, 0\} + \hat{\beta} \mathbb{E}[V(k', z') | z] \right\},$$

where

$$A(k', k, z; w, \theta) = (1 - \tau_{\text{corp}})(1 - \alpha_l) y(k, z; w) + \tau_{\text{corp}} \delta k - I - (1 - \tau_{\text{corp}} \phi) \Phi(k', k).$$

Numerically this is what `GourioMiao2010.m` solves: it sets $n_d = 0$, gives the toolkit a single decision grid (capital), and never discretizes dividends. The optimal $d^* = \max\{A, 0\}$ and $s^* = \max\{-A, 0\}$ are recovered from k' -policy at the post-processing stage for table-building (see `GourioMiao2010_DividendFn.m` and `GourioMiao2010_NewEquityFn.m`).

Financing regimes. Because exactly one of d^* , s^* is non-zero, the three GM2010 financing regimes correspond cleanly to the sign of A :

$$\underbrace{A > 0}_{\text{dividend regime}} \quad \underbrace{A \approx 0}_{\text{liquidity-constrained}} \quad \underbrace{A < 0}_{\text{equity-issuance regime}}.$$

The liquidity-constrained mass concentrates at the kink in F where the slope in k' changes; with a discrete a -grid the optimum sits on a grid point, so the regime is identified with $|A| \leq \text{tol}$ (we use the a -grid spacing).

4 Calibration

The baseline parametrization (Figure 1) matches the values reported in GM2010.

Productivity is discretized with a 10-point Tauchen–Hussey grid; capital uses $n_a = 701$ evenly spaced points on $[0.001, 8]$.

5 Baseline results

With the wage solved from the household labor-market FOC, Figures 2 and 3 report the aggregate and cross-sectional moments and the share of firms in each financing regime, each alongside the corresponding original.

6 Dividend tax reform

The reform exercise considers four pairs of (τ_d, τ_{cg}) relative to the baseline. Figure 4 gives the long-run percentage changes in aggregates; Figure 5 reports how the firm distribution across financing regimes shifts under the small $(\tau_d = 0.22)$ reform; Figure 6 decomposes the productivity gain.

7 Sensitivity

Figures 7 and 8 re-run the baseline moments and the $(\tau_d, \tau_{cg}) = (0.15, 0.15)$ reform for alternative values of ρ_z , σ_z , and the adjustment-cost parameter ϕ^c .

TABLE 3—BASELINE PARAMETRIZATION

	Parameter	Value
Corporate income tax	τ_c	0.340
Personal income tax	τ_i	0.250
Dividend tax	τ_d	0.250
Capital gains tax	τ_g	0.200
Exponent on capital	α_k	0.311
Exponent on labor	α_l	0.650
Shock persistence	ρ	0.767
Shock standard deviation	σ	0.211
Depreciation rate	δ	0.095
Discount factor	β	0.971
Weight on leisure	h	6.616
Adjustment cost	ψ	1.080

Baseline Parametrization		
	Parameter	Value
Corporate income tax	τ_{corp}	0.340
Personal income tax	τ_i	0.250
Dividend tax	τ_d	0.250
Capital gains tax	τ_{cg}	0.200
Exponent on capital	α_k	0.311
Exponent on labor	α_l	0.650
Shock persistence	ρ	0.767
Shock standard deviation	σ	0.211
Depreciation rate	δ	0.095
Discount factor	β	0.971
Weight on leisure	h	6.616
Adjustment cost	ϕ	1.080

Figure 1: Baseline parametrization. Top: original from Gourio & Miao (2010). Bottom: replication.

8 Notes on the code

- `GourioMiao2010_ReturnFn.m` implements the simplified return function $F = \frac{1-\tau_d}{1-\tau_{cg}} \max\{A, 0\} + \min\{A, 0\}$, with y and π in closed form. The driver sets $n_d = 0$ and $d_grid = []$.
- `GourioMiao2010_DividendFn.m` and `GourioMiao2010_NewEquityFn.m` return $d^* = \max\{A, 0\}$ and $s^* = \max\{-A, 0\}$ respectively; both are used only at the post-processing stage to build tables.
- Regime classification uses `tol` equal to the a -grid spacing, applied to d^* and s^* ; no separate tolerance on dividends is needed because d^* and s^* are computed analytically.

TABLE 4—AGGREGATE AND CROSS-SECTIONAL MOMENTS IN THE BASELINE MODEL

Variable	Data	Model
Investment rate	0.095	0.095
Aggregate dividends/earnings	0.137	0.353
Aggregate new equity/investment	0.130	0.148
Volatility of investment rate	0.156	0.156
Autocorrelation of investment rate	0.596	0.648
Volatility of earnings/capital	0.623	0.175
Autocorrelation of earnings/capital	0.791	0.658

Notes: The investment rate is computed from the National Income Accounts, and the other data are computed using COMPUSTAT. See Appendix Section B for the variable definition. Model moments are computed using parameter values listed in Table 3.

Aggregate and Cross-Sectional Moments in the Baseline Model

Variable	Data	Model
I/Y	-	0.152
Investment rate (I/K)	-	0.095
Aggregate dividends/earnings	-	0.350
Aggregate new equity/investment	-	0.139
Volatility of investment rate	-	0.155
Autocorrelation of investment rate	-	0.649
Volatility of earnings/capital	-	0.175
Autocorrelation of earnings/capital	-	0.658

Notes: Autocorrelations are computed exactly from the stationary distribution and the (kprime,z) transition (no simulation). Volatility is the cross-sectional standard deviation under the stationary distribution. From codes of GM2010 it is clear that Investment Rate is actually reporting I/K, not I/Y. I include I/Y as reading the paper this is what I had initially assumed it to be.

Figure 2: Aggregate and cross-sectional moments. Top: original from Gourio & Miao (2010). Bottom: replication.

TABLE 5—DISTRIBUTION OF FIRMS ACROSS FINANCE REGIMES IN THE BASELINE MODEL

	Equity issuance regime	Liquidity constrained regime	Dividend distribution regime
Share of firms	0.201	0.342	0.457
Share of capital	0.108	0.229	0.663
Share of investment	0.402	0.436	0.161
Earnings-capital ratio	0.431	0.264	0.170
Investment-capital ratio	0.354	0.181	0.023
Average Tobin's q	2.633	1.941	1.348

Notes: The share of firms for a regime is equal to the total number of firms in that regime divided by the total number of firms in all regimes. The share of capital (investment) for a regime is equal to the total capital stock (resp. investment) of the firms in that regime divided by aggregate capital stock (investment) of all firms. The earnings-capital ratio for a regime is equal to the earnings of the firms in that regime divided by their total capital stock, i.e., it is capital-weighted. The investment-capital ratio and Tobin's q are computed in a similar way. Parameter values are listed in Table 3.

Distribution of Firms across Finance Regimes in the Baseline Model

	Equity issuance regime	Liquidity constrained regime	Dividend distribution regime
Share of firms	0.197	0.355	0.448
Share of capital	0.110	0.215	0.675
Share of investment	0.416	0.383	0.201
Earnings-Capital Ratio	0.444	0.240	0.176
Investment-Capital Ratio	0.358	0.169	0.028
Average Tobins q	2.640	1.957	1.359

Notes: The share of firms for a regime is equal to the total number of firms in that regime divided by the total number of firms in all regimes. The share of capital (investment) for a regime is equal to the total capital stock (resp. investment) of the firms in that regime divided by aggregate capital stock (investment) of all firms. The earnings-capital ratio for a regime is equal to the earnings of the firms in that regime divided by their total capital stock, i.e., it is capital-weighted. The investment-capital ratio and Tobins q are computed in a similar way.

Figure 3: Distribution of firms across finance regimes. Top: original from Gourio & Miao (2010). Bottom: replication.

TABLE 6—AGGREGATE EFFECTS OF THE DIVIDEND TAX REFORM IN THE BASELINE MODEL

	$\tau_d = 0.22$ $\tau_g = 0.20$	$\tau_d = 0.20$ $\tau_g = 0.20$	$\tau_d = 0.15$ $\tau_g = 0.15$	$\tau_d = 0$ $\tau_g = 0$
Capital	0.27	0.52	4.26	13.95
Output	0.58	1.00	2.15	5.04
Consumption	0.38	0.63	1.30	2.91
Dividends	6.23	N/A	N/A	N/A
Equity issuance	40.05	N/A	N/A	N/A
Wage	0.48	0.82	1.72	3.95
Firm value	3.40	5.78	10.52	24.09
Welfare	0.31	0.52	1.04	2.27

Notes: When we change the dividend and capital gains tax rates, we fix all other parameter values as in Table 3. All results are measured in percentage change from the initial steady state before the reform.

Aggregate Effects of the Dividend Tax Reform in the Baseline Model

	$\tau_d = 0.22$ $\tau_{cg} = 0.20$	$\tau_d = 0.20$ $\tau_{cg} = 0.20$	$\tau_d = 0.15$ $\tau_{cg} = 0.15$	$\tau_d = 0$ $\tau_{cg} = 0$
Capital	0.22	0.57	4.14	13.72
Output	0.56	1.01	2.09	4.94
Consumption	0.36	0.64	1.26	2.84
Dividends	6.66	N/A	N/A	N/A
Equity Issuance	44.22	N/A	N/A	N/A
Wage	0.46	0.82	1.69	3.89
Firm Value	3.41	5.76	10.52	24.15
Welfare	0.30	0.52	1.02	2.23

Notes: All results are measured in percentage change from the initial steady state before the reform. When dividend tax and capital gains tax are equal, firms are indifferent between dividends and equity issuance, hence the N/A.

Figure 4: Aggregate effects of the dividend tax reform. Top: original from Gourio & Miao (2010). Bottom: replication.

TABLE 7—DISTRIBUTION OF FIRMS ACROSS FINANCE REGIMES FOR $\tau_d = 0.22$ and $\tau_g = 0.20$

	Equity issuance regime	Liquidity constrained regime	Dividend distribution regime
Share of firms	0.248	0.259	0.493
Share of capital	0.138	0.169	0.693
Share of investment	0.517	0.325	0.158
Earnings-capital ratio	0.415	0.264	0.171
Investment-capital ratio	0.357	0.183	0.022
Average Tobin's q	2.620	2.001	1.406

Notes: The share of firms for a regime is equal to the total number of firms in that regime divided by the total number of firms in all regimes. The share of capital (investment) for a regime is equal to the total capital stock (resp. investment) of the firms in that regime divided by aggregate capital stock (investment) of all firms. The earnings-capital ratio for a regime is equal to the earnings of the firms in that regime divided by their total capital stock, i.e., it is capital-weighted. The investment-capital ratio and Tobin's q are computed in a similar way. Except for τ_d and τ_g , other parameter values are listed in Table 3.

Distribution of Firms across Finance Regimes for $\tau_d = 0.22$ and $\tau_{cg} = 0.20$

	Equity issuance regime	Liquidity constrained regime	Dividend distribution regime
Share of firms	0.261	0.254	0.485
Share of capital	0.162	0.132	0.706
Share of investment	0.581	0.213	0.206
Earnings-Capital Ratio	0.410	0.211	0.178
Investment-Capital Ratio	0.341	0.153	0.028
Average Tobins q	2.532	2.032	1.420

Notes: The share of firms for a regime is equal to the total number of firms in that regime divided by the total number of firms in all regimes. The share of capital (investment) for a regime is equal to the total capital stock (resp. investment) of the firms in that regime divided by aggregate capital stock (investment) of all firms. The earnings-capital ratio for a regime is equal to the earnings of the firms in that regime divided by their total capital stock, i.e., it is capital-weighted. The investment-capital ratio and Tobins q are computed in a similar way.

Figure 5: Distribution of firms across finance regimes after the small reform. Top: original from Gourio & Miao (2010). Bottom: replication.

TABLE 8—PRODUCTIVITY GAINS FROM THE DIVIDEND TAX CUT

	$\tau_d = 0.25$	$\tau_d = 0.22$	$\tau_d = 0.20$
Percentage change in <i>TFP</i>	0.00	0.479	0.721
Percentage change in <i>Y/L</i>	0.00	0.430	0.818
Correlation between $\ln k$ and $\ln z$	0.438	0.450	0.457
Regression coefficient of $\ln k$ on $\ln z$	1.203	1.268	1.312

Note: When we change the dividend tax rate, we fix all other parameter values as in Table 3.

Productivity Gains from the Dividend Tax Cut

	$\tau_d = 0.25$	$\tau_d = 0.22$	$\tau_d = 0.20$
Percentage change in TFP	0.000	0.423	0.709
Percentage change in <i>Y/L</i>	0.000	0.459	0.821
Correlation between $\ln(k)$ and $\ln(z)$	0.437	0.451	0.460
Regression coefficient of $\ln(k)$ on $\ln(z)$	1.194	1.261	1.305

Figure 6: Productivity gains from the dividend tax cut. Top: original from Gourio & Miao (2010). Bottom: replication.

TABLE 9—MOMENTS FOR DIFFERENT PARAMETER VALUES

	Data	$\rho = 0.65$	$\rho = 0.85$	$\sigma = 0.1$	$\sigma = 0.3$	$\psi = 0.5$	$\psi = 1.5$
Aggregate dividends/earnings	0.137	0.290	0.435	0.296	0.413	0.481	0.315
Aggregate new equity/investment	0.130	0.007	0.340	0.010	0.294	0.419	0.069
Volatility of investment rate	0.156	0.098	0.224	0.076	0.208	0.254	0.127
Autocorrelation of investment rate	0.596	0.566	0.683	0.690	0.603	0.588	0.668
Volatility of earnings/capital	0.623	0.160	0.176	0.080	0.273	0.157	0.186
Autocorrelation of earnings/capital	0.791	0.557	0.732	0.720	0.588	0.647	0.662

Note: When we change one parameter value, we fix other parameter values as in Table 3.

Moments for Different Parameter Values

	Data	$\rho = 0.65$	$\rho = 0.85$	$\sigma = 0.1$	$\sigma = 0.3$	$\phi = 0.5$	$\phi = 1.5$
Aggregate dividends/earnings	-	0.288	0.433	0.294	0.411	0.477	0.312
Aggregate new equity/investment	-	0.003	0.331	0.006	0.285	0.407	0.063
Volatility of investment rate	-	0.098	0.221	0.076	0.207	0.252	0.126
Autocorrelation of investment rate	-	0.567	0.692	0.692	0.604	0.591	0.669
Volatility of earnings/capital	-	0.160	0.178	0.079	0.275	0.158	0.185
Autocorrelation of earnings/capital	-	0.557	0.732	0.720	0.588	0.647	0.662

Figure 7: Moments for different parameter values. Top: original from Gourio & Miao (2010). Bottom: replication.

TABLE 10—SENSITIVITY ANALYSIS OF DIVIDEND TAX REFORM FOR DIFFERENT PARAMETER VALUES

	Capital	Output	Consumption	Wage
Baseline	4.26	2.15	1.30	1.72
$\rho = 0.65$	3.58	1.34	0.78	1.08
$\rho = 0.85$	5.40	2.32	1.38	1.86
$\sigma = 0.1$	3.57	1.48	0.88	1.19
$\sigma = 0.3$	5.36	2.34	1.39	1.88
$\psi = 0.5$	5.72	2.85	1.72	2.28
$\psi = 1.5$	4.06	1.89	1.15	1.50

Notes: When we change one parameter value, we fix other parameter values as in Table 3. All results are measured in percentage change from the initial steady state.

Moments for Different Parameter Values				
	Capital	Output	Consumption	Wage
Baseline	4.16	2.11	1.27	1.69
$\rho = 0.65$	3.66	1.36	0.79	1.11
$\rho = 0.85$	5.30	2.29	1.37	1.81
$\sigma = 0.1$	3.55	1.46	0.86	1.18
$\sigma = 0.3$	5.11	2.26	1.35	1.80
$\phi = 0.5$	5.61	2.80	1.69	2.24
$\phi = 1.5$	3.89	1.81	1.10	1.45

Notes: All results are measured in percentage change of model with tax-reform from model before tax-reform.

Figure 8: Reform effects under different parameter values. Top: original from Gourio & Miao (2010). Bottom: replication.